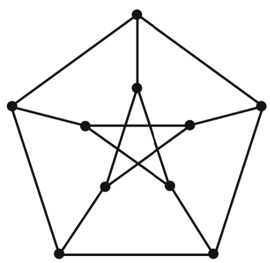
Home assignment: day 12 → day 15

Deadline: 2026-07-17, 16:59.

1. A convex polyhedron has 20 triangular faces and 12 pentagonal faces. How many vertices and edges does it have?
2. Prove that $K_{3,3}$ is non-planar.
3. Using Kuratowski's theorem or otherwise check whether the Petersen graph is planar.



4. Watch 3Blue1Brown video «Why this puzzle is impossible»,

   https://www.youtube.com/watch?v=VvCytJvd4H0&t=36s, .

Demo for the quiz on day 13

1. A polyhedron with 1 hole has 30 vertices and 35 edges. How many faces does it have?
2. Draw $K_{1,3,4}$. Using Kuratowski's theorem or otherwise check whether $K_{1,3,4}$ is planar.